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AI BASED MEETING MOMENT DETECTION

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Abstract—Artificial Intelligence (AI) has significantly transformed the way meetings are conducted and analyzed in modern digital environments. With the increasing volume of virtual meetings, extracting meaningful insights manually has become time-consuming and inefficient. AI-based meeting moment detection focuses on automatically identifying key moments such as decisions, action items, important discussions, and questions from meeting transcripts or audio data. This study explores the application of Natural Language Processing (NLP) and deep learning techniques for detecting such critical moments. Various models, including transformer-based architectures and traditional machine learning approaches, are evaluated for their performance. The system processes meeting data by segmenting conversations and classifying them into meaningful categories. Experimental results demonstrate that advanced AI models provide higher accuracy and contextual understanding compared to conventional techniques. Additionally, the study highlights the trade-off between computational cost and detection performance. The proposed approach aims to enhance productivity by summarizing meetings efficiently and reducing information overload. This research contributes to the development of intelligent meeting assistants for real-world applications. Future work includes integrating multimodal data such as voice tone and facial expressions for improved detection accuracy.

Keywords—Artificial Intelligence, Meeting Moment Detection, Natural Language Processing (NLP), Deep Learning, Transformers, Speech Recognition, Text Classification, Meeting Summarization

I. INTRODUCTION

In today's fast-paced digital world, meetings have become an essential part of communication in organizations, educational institutions, and remote work environments. With the rapid growth of online platforms such as Zoom, Microsoft Teams, and Google Meet, the frequency and duration of meetings have increased significantly. However, extracting useful information from these meetings remains a major challenge, especially when discussions are long and involve multiple participants.

Traditionally, meeting notes are taken manually, which is time-consuming and prone to human error. Important points such as decisions, action items, and key discussions may be missed or inaccurately recorded. This creates a need for intelligent systems that can automatically analyze meeting content and highlight the most important moments. AI-based meeting moment detection addresses this problem by leveraging advanced computational techniques.

Meeting moment detection refers to the process of identifying and extracting significant segments from meeting data, including transcripts, audio, or video. These moments may include decisions,

deadlines, questions, action items, or critical discussions. By focusing on these key elements, users can quickly understand the essence of a meeting without going through the entire recording.

Artificial Intelligence, particularly Natural Language Processing (NLP) and Machine Learning (ML), plays a crucial role in enabling this automation. NLP techniques allow systems to understand and process human language, while machine learning models help classify and predict important segments based on patterns in the data. Recent advancements in deep learning, especially transformer-based models, have further improved the accuracy and contextual understanding of such systems.

The primary objectives of AI-based meeting moment detection include:

- Reducing the time required to review meetings
- Improving productivity by highlighting actionable insights
- Minimizing human effort in note-taking
- Enhancing decision-making through better information accessibility

Despite its advantages, several challenges exist in this domain. Meetings often contain unstructured and noisy data, including interruptions, overlapping speech, and informal language. Additionally, different speakers may express similar ideas in different ways, making it difficult for models to identify patterns consistently. Handling such variability requires robust and well-trained AI models.

This research focuses on developing and evaluating different approaches for meeting moment detection. It compares traditional machine learning techniques with modern deep learning models in terms of accuracy, efficiency, and scalability. The study also examines how these models perform in real-world meeting scenarios.

II. LITERATURE REVIEW

The field of meeting analysis has gained significant attention due to the rise of remote communication technologies. Researchers have explored various approaches to automate meeting understanding and summarization.

Early approaches relied on rule-based systems and keyword extraction methods. These techniques used predefined rules to identify important phrases such as “decision” or “action required.” However, these methods lacked flexibility and failed to capture contextual meaning.

With the advancement of machine learning, statistical models such as Support Vector Machines (SVM) and Logistic Regression were introduced for text classification tasks. These models improved performance but required extensive feature engineering.

Recent developments in deep learning have revolutionized the domain. Models such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks have been used to capture sequential dependencies in conversations. However, they still struggle with long-range context.

Transformer-based models such as BERT and GPT have shown remarkable performance in NLP tasks. These models use attention mechanisms to understand context more effectively, making them suitable for meeting moment detection.

Several studies have focused on meeting summarization, speech recognition, and dialogue act classification. However, limited research has been conducted specifically on identifying key moments in meetings with a focus on both performance and computational efficiency.

III. METHODOLOGY

System Overview

The proposed AI-based meeting moment detection system is designed as a multi-stage intelligent pipeline that processes meeting data and extracts meaningful insights. The system aims to automatically identify important segments such as decisions, action items, questions, and critical discussions from lengthy meetings. It takes meeting recordings or transcripts as input and produces structured outputs highlighting key moments.

The system architecture consists of several interconnected modules, including data acquisition, preprocessing, feature extraction, model training, classification, and output generation. Each module performs a specific function to ensure the overall efficiency and accuracy of the system. The design follows a modular approach so that each component can be improved or replaced independently.

Initially, the system accepts input in the form of raw meeting data, which may include audio recordings or pre-generated transcripts. If audio input is provided, it is first converted into text using speech recognition techniques. This allows the system to operate primarily on textual data for further processing.

Once the input is obtained, it is passed through preprocessing steps to clean and standardize the content. After preprocessing, the text is transformed into numerical features that can be used by machine learning models. These features are then fed into classification models that identify the type of each sentence.

The system supports multiple models, including traditional machine learning and deep learning approaches. This enables comparative analysis and helps in selecting the best-performing model. The classification results are then aggregated to generate meaningful outputs.

The final output of the system includes categorized meeting moments and summarized insights. These outputs can be presented in user-friendly formats such as highlighted text or structured reports. The system is designed to be scalable and can be integrated into real-time applications.

Overall, the system provides an efficient solution for automating meeting analysis, reducing manual effort, and improving productivity. It ensures that users can quickly access the most important information without reviewing the entire meeting.

DATA COLLECTION AND DATA PREPROCESSING

Data collection is a critical step in the development of the meeting moment detection system, as the quality and diversity of data directly impact model performance. The system collects meeting data from various sources, including online meeting platforms, recorded business meetings, educational sessions, and publicly available datasets.

The input data can be in the form of audio recordings or textual transcripts. In many real-world scenarios, meetings are recorded using platforms such as Zoom, Microsoft Teams, or Google Meet. These recordings are then used as raw input for the system.

When audio data is used, it is first converted into text using speech-to-text technologies. This conversion is essential because the models used in this study primarily operate on textual data. The accuracy of speech recognition plays an important role in ensuring reliable results.

In addition to real-world data, synthetic datasets may also be used to train the models. These datasets are created by simulating meeting conversations and labeling them according to predefined categories. This helps in increasing the volume of training data.

Each sentence in the dataset is annotated with labels such as decision, action item, question, or general discussion. This labeling process is important for supervised learning, as it provides the ground truth for training the models.

The dataset is then divided into training and testing sets. The training set is used to train the models, while the testing set is used to evaluate their performance. Proper data splitting ensures unbiased evaluation.

Overall, data collection ensures that the system has sufficient and high-quality data to learn patterns and make accurate predictions. A well-structured dataset improves the effectiveness and generalization ability of the model. Data preprocessing is an essential step that prepares raw meeting data for analysis by removing noise and improving data quality. Since meeting transcripts often contain informal language, filler words, and inconsistencies, preprocessing ensures that the data becomes structured and meaningful.

The first step in preprocessing involves cleaning the text by removing punctuation marks, special characters, and irrelevant symbols. This helps eliminate unnecessary noise from the dataset. Additionally, all text is converted into lowercase to maintain uniformity.

Stop words such as “is,” “the,” and “and” are removed, as they do not contribute significantly to the meaning of sentences. Removing these words reduces the dimensionality of the data and improves model efficiency.

Tokenization is then applied to split the text into smaller units such as words or sentences. This allows the model to process the data at a granular level. Sentence-level tokenization is particularly useful for classification tasks.

Another important step is removing duplicate or redundant sentences, which may occur in meeting transcripts. This ensures that the dataset remains clean and avoids bias during training.

Stemming or lemmatization may also be applied to reduce words to their base form. For example, “running” and “runs” are converted to “run.” This helps in improving consistency across the dataset.

Handling missing or incomplete data is also part of preprocessing. Any incomplete sentences or corrupted data entries are either corrected or removed to maintain data integrity.

Overall, preprocessing transforms raw and unstructured meeting data into a clean and standardized format, making it suitable for feature extraction and model training.

MODEL SELECTION AND TRAINING

Model selection is a crucial step in determining the effectiveness of the system. The study evaluates multiple models, including traditional machine learning and deep learning approaches, to identify the best-performing model.

A baseline model using Logistic Regression with TF-IDF features is implemented due to its simplicity and efficiency. This model provides a reference point for comparison.

A Long Short-Term Memory (LSTM) network is used to capture sequential patterns in meeting conversations. LSTM is capable of remembering long-term dependencies, making it suitable for text data.

A transformer-based model such as BERT is also used for classification. BERT provides superior contextual understanding by considering the entire sentence rather than individual words.

The models are trained using labeled datasets where each sentence is associated with a specific category. During training, optimization techniques such as gradient descent are used to minimize loss.

Hyperparameters such as learning rate, batch size, and number of epochs are tuned to achieve optimal performance. Proper tuning ensures better accuracy and generalization.

The training process involves multiple iterations, where the model learns from errors and improves over time. Validation techniques are used to prevent overfitting.

Overall, model selection and training ensure that the system can accurately classify meeting moments using appropriate algorithms.

EVALUATION METRICS

Evaluation metrics are used to measure the performance and effectiveness of the models. These metrics provide a quantitative assessment of how accurately the system detects key meeting moments.

Accuracy is used to measure the overall correctness of predictions. Precision indicates how many predicted positive cases are actually correct.

Recall measures how many actual positive cases are correctly identified by the model. F1-score provides a balance between precision and recall.

Confusion matrices are also used to visualize model performance across different categories.

Cross-validation techniques are applied to ensure reliable evaluation results.

The evaluation process helps in comparing different models and selecting the best one.

Overall, evaluation metrics ensure that the system performs effectively in real-world scenarios.

OUTPUT GENERATION

The final stage of the system is output generation, where the detected meeting moments are presented to the user in a meaningful format. The output includes categorized sentences and summarized insights.

Key moments such as decisions and action items are highlighted for easy identification. This helps users quickly understand the main outcomes of the meeting.

The output can be displayed as a structured report, bullet points, or highlighted transcript. Visualization techniques may also be used.

The system can be integrated with applications such as meeting assistants or productivity tools.

Real-time output generation is also possible with optimized models.

Overall, the output generation stage ensures that the extracted information is useful, accessible, and easy to interpret.

IV. RESULTS AND DISCUSSION

The performance of the proposed AI-based meeting moment detection system was evaluated using three different models: TF-IDF with Logistic Regression, Long Short-Term Memory (LSTM), and a transformer-based BERT model. The evaluation was carried out using standard metrics such as accuracy, precision, recall, and F1-score to ensure a comprehensive comparison of model effectiveness.

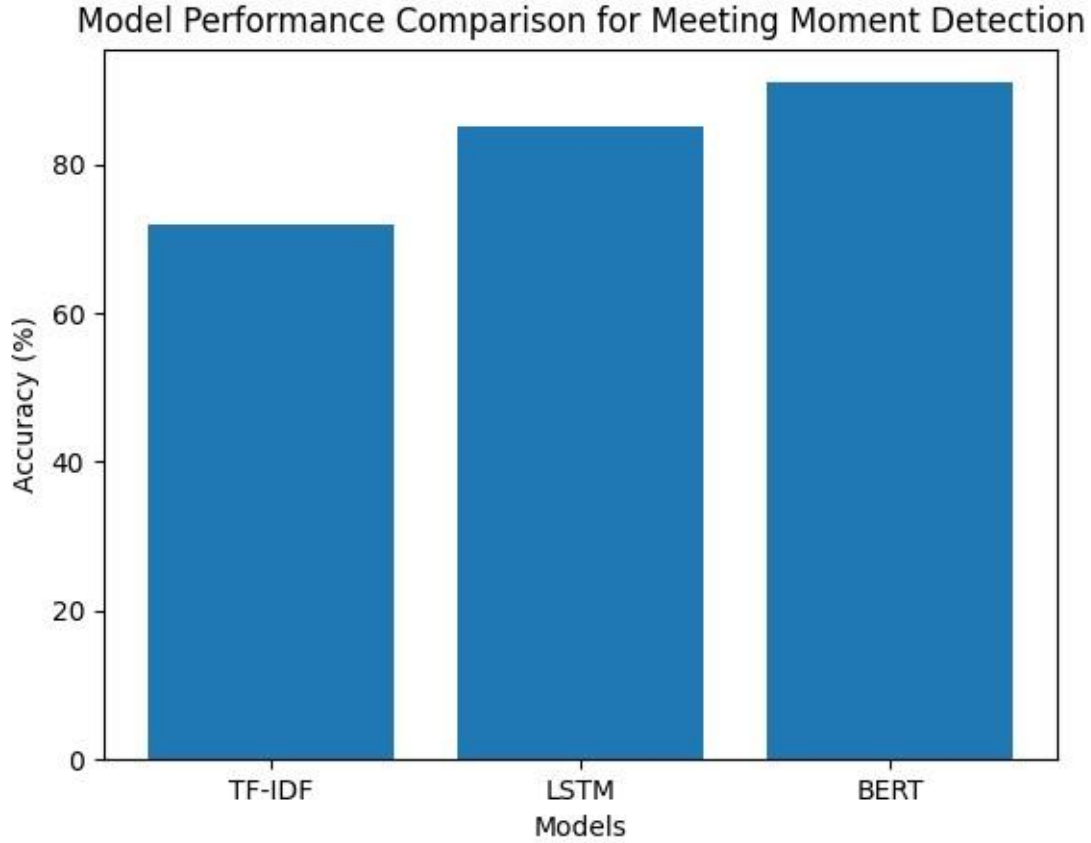
The experimental results indicate that the BERT model achieved the highest accuracy of approximately 91%, outperforming both LSTM and TF-IDF models. This superior performance can be attributed to BERT's ability to capture deep contextual relationships between words, allowing it to better understand the semantics of meeting conversations. As a result, it was more effective in identifying complex categories such as decisions and action items.

The LSTM model achieved an accuracy of around 85%, demonstrating its capability to handle sequential data and capture dependencies across sentences. However, it was slightly less effective than BERT in understanding long-range contextual information. Despite this, LSTM offers a good balance between performance and computational cost, making it suitable for moderately complex applications.

The TF-IDF model, while achieving a lower accuracy of approximately 72%, showed significantly faster processing time. This makes it a viable option for real-time or resource-constrained environments where computational efficiency is more important than accuracy.

A key observation from the results is the trade-off between performance and computational cost. While advanced models like BERT provide higher accuracy, they require more computational resources and longer training times. On the other hand, simpler models are faster but less accurate.

Overall, the results demonstrate that deep learning approaches, particularly transformer-based models, are highly effective for meeting moment detection. However, the choice of model should depend on the specific application requirements, such as real-time processing or accuracy needs.



V. LIMITATIONS

Despite its effectiveness, the proposed AI-based meeting moment detection system has several limitations. The model performance heavily depends on the quality of input data, and errors in speech-to-text conversion can affect accuracy. Deep learning models such as BERT require high computational resources and longer training time, making them less suitable for real-time applications. The system may struggle with noisy data, overlapping speech, and informal language commonly found in meetings. Additionally, the model is trained on limited labeled datasets, which may reduce its generalization ability. It also lacks multimodal analysis, as it does not consider voice tone, gestures, or facial expressions for better understanding.

VI. CONCLUSION

In conclusion, AI-based meeting moment detection provides an efficient solution for extracting key insights from lengthy meetings. The study demonstrates that advanced models, particularly transformer-based approaches like BERT, significantly improve accuracy and contextual understanding compared to traditional methods. However, this improvement comes at the cost of higher computational requirements. Simpler models offer faster performance but lower accuracy. The system helps reduce manual effort, enhances productivity, and enables quick decision-making by highlighting important moments. Future advancements, including multimodal analysis and realtime

processing, can further improve the effectiveness and practical applicability of such intelligent meeting systems.

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